

Two cracks in the answer key — and where we're borrowing the fix from

A colleague sat down and manually re-checked the loan checker's homework against a real loan. She found two real problems with how the "answer key" (the compiled checklist) gets written. This is what she found, in plain English, and the plan for fixing both — plus proof that other, much bigger systems already solved the second problem the same way we're about to.

Picture a teacher's answer key for a 800-question test. Two very different things can go wrong with an answer key that have nothing to do with the student's actual answers:

1. A question's answer key has the direction backwards — like a key that says "correct if the essay is UNDER 500 words" when the actual assignment was "correct if the essay is 500 words or more." The grader would mark a perfectly good 500-word essay wrong.

2. A question gets asked of every student, even the ones it doesn't apply to — like grading every single paper on "did you cite your extra-credit reading correctly," even for students who never did the extra credit. That question should say "N/A" for them, not "you forgot something."

The answer key had a "greater than" pointing the wrong way

One real rule: "a loan needs mortgage insurance once it's borrowed more than 80% of the home's value." That's a fail-condition sentence — it describes when the loan is in trouble. Our AI reader copied the comparison word literally instead of flipping it around to describe the pass condition the computer actually checks against.

WHAT GOT WRITTEN DOWN (WRONG)

Pass only if the loan borrowed **MORE THAN** 80% of the home's value.

WHAT IT SHOULD SAY

Pass if the loan borrowed **80% OR LESS** of the home's value.

A real loan borrowed **exactly** 80% — right on the line. With the backwards answer key, the computer said "fail, needs mortgage insurance" — the opposite of the true rule. We checked every question of this shape and found **45 more**, out of 495 total, with the same kind of backwards answer.

Questions got asked even when they don't apply to this loan

One real rule: "if the borrower used a cash gift from a relative to help make the down payment, there must be a signed gift letter on file." On the loan our colleague checked, there was **no gift** — the buyer used their own savings. The rule should have said "not applicable — no gift here." Instead, the loan checker flagged it as an unresolved problem, because nothing in the system knows to first ask "does this loan even involve a gift?" before grading the gift-letter question.

This is the deeper issue: **the checklist has no way to say "skip this question unless X is true about the loan."** Every one of the 8,000+ questions runs on every loan today, whether it makes sense for that loan or not.

How we're fixing both — the updated end-to-end picture

Same four stages as before (turn the rulebook into a checklist, read the loan's paperwork, run the checklist, file the results) — with two small, targeted additions.

① Turn the rulebook into a

8,442 ru

AI writes each
a check – and
'only if...' w
it only applie
some loans (FI

Double-checked: (

PASS/FAIL point
right way? (FIX

Signed-c
checkli

② Read this loan's pa

Closing pap
+ lender f.

Facts about 1
loan – incl.
a gift use

③ Run the checklist on 1

Each
questi

tag match
this loar

no

yes

N/A

graded:
PASS / FAI
FLAG

④ File the f

One clean repor
nothing backwar
nothing irrelev
flagged

We're not guessing at this fix — two independent proofs it's the right shape

The "only ask this if..." tag (Problem 2's fix) isn't a new invention. We went looking for how other serious systems handle the exact same "does this rule even apply?" question, and found it solved the same way in two very different places.

GOVERNMENT & ENTERPRISE ACCESS-CONTROL RULES

The standard used to decide "is this person even allowed to try this action" (used across banking, government, and healthcare IT security) has kept a rule's "does this apply to this case" check completely separate from its "what's the actual decision" check for over 20 years — and gives "doesn't apply" its own clean answer, never a maybe.

A SIBLING DEMO WE ALREADY BUILT, FOR THIS SAME BANK

A separate scenario-modeling tool one of our own AI engineers built — pulling real PennyMAC and Fannie Mae guideline pages into its own checklist — already tags individual rules with "only applies if..." conditions, kept separate from the rule's actual pass/fail math. We found this independently, then confirmed our fix should work the same way.

One important guardrail, kept from day one: the "only ask this if..." tag can only ever be based on what the original rule's own wording already says. It is never allowed to invent a new condition using outside knowledge (like a Fannie Mae guidebook) — that guardrail is what keeps the checklist honest: it can help us understand an existing rule better, but it can never quietly add a new one nobody asked for.

Where this stands right now

- ✓

Both problems confirmed on a real loan
- ✓

Fix plans written up, in detail
- ✓

Design double-checked against two outside examples
- ⌚

Not built yet — waiting on review before any code changes

